

TUSHAR SAINI

Full-Stack / Backend Engineer · Targeting SDE-2

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WORK EXPERIENCE

Software Engineer · Innovaccer · Noida, India Dec 2024 – Present

- ▶ **Clinical Workflow Pipelines:** Authored HLD and technical spec docs for internal stakeholders; architected async provider & coder validation pipelines for **500+ providers** — orchestrating distributed execution, retry handling, and payer compliance verification across medical billing. Reduced manual review by **40%**, claim rejection rate by **35%**. *[Python, FastAPI, Redis, arq, MongoDB, Elasticsearch, React]*
- ▶ **RPA Automation Bridge:** Designed a browser-native RPA bridge (Innote ↔ athenahealth EHR) using AES-encrypted WebSocket channel via Chrome extension — automating claim updates for **2,000+ daily encounters** and eliminating **8+ manual steps** per workflow with **zero human intervention**. *[React.js, WebSockets, Chrome Extension APIs]*
- ▶ **VBC Platform:** Owned a Value-Based Care platform contributing **\$5M+ ARR** — drove cross-team collaboration with product, QA, and clinical ops to deliver for **50,000+ patients** across **10+ care programs** with patient-specific guideline execution and multi-EHR order integration via a config-driven abstraction layer. *[React, Python, FastAPI]*
- ▶ **Medication Interaction Engine:** Architected a distributed drug validation engine processing **400k+ validations/month**, sustaining **5 QPS (10 QPS peak)** at **P95 <180ms** under production load — preventing an estimated **1,200+ adverse drug events/month**. *[Python, FastAPI, Redis]*
- ▶ **Distributed Rate Limiter:** Designed a Redis Sorted Sets + Lua-based throttling system enforcing atomic rate limiting across **12+ microservices** at **80,000+ inter-service requests/hour** — **zero race conditions**, sub-millisecond enforcement overhead. *[Python, FastAPI, Redis, Lua]*
- ▶ **Real-Time Mobile Platform:** Led mobile engineering for a doctor-patient collaboration platform serving **10,000+ MAU** — reduced crash rate by **80%**, cut redundant API calls by **30%**, saving **~15,000 unnecessary requests/day**. *[Flutter, Dart, WebSockets, OAuth2.0, RBAC]*

Software Engineer · Niyo · Bangalore, India Jan 2023 – Dec 2024

- ▶ **Bank Onboarding Platform:** Led end-to-end development of SBM Bank's digital onboarding, coordinating with Spense APIs (SBM's integration partner) — **60,000+ activations in 90 days**, improved KYC completion rate by **38%** via optimised step sequencing and error recovery. *[Flutter, Dart, JavaScript, Webpack, PWA]*
- ▶ **Notification Center:** Led end-to-end design and delivery of a full-stack notification system for **500,000+ users** — built UI notification center with in-app deep-link routing, click tracking, and grouped event processing, improving engagement from **4% → 9% (+125% lift)**. *[Flutter, Dart, Kotlin, Swift]*
- ▶ **Serverless Notification Pipeline:** Productionized an event-driven AWS Lambda push pipeline processing **10,000+ notifications/day** at **99.4% delivery success** — up from ~91% with the prior synchronous architecture. *[Node.js, AWS Lambda]*
- ▶ **App-Rater Workflow API:** Engineered a rating workflow API at **sub-40ms P99 latency**, collecting **50,000+ in-app ratings** and driving a **+0.3-point improvement** in app store rating. *[Node.js, Nest.js, Firebase]*

PROJECT

Local RAG Pipeline for Large-Scale Document QA LlamaIndex · Qdrant · Ollama · Langfuse · Celery · Redis

- ▶ **High-Scale Ingestion:** Built a distributed batch ingestion pipeline using Celery and Redis to parse (Docling) and semantically chunk (LlamaIndex) **10M+ PDF and text documents** asynchronously with zero memory overflow.
- ▶ **Hybrid Vector Search:** Implemented hybrid search combining dense vector retrieval (mxbai-embed-large) and BM25 sparse retrieval in Qdrant; integrated BGE-Reranker to re-score top-k retrieved chunks, significantly improving answer precision.
- ▶ **Fully Private Inference:** Deployed phi4 via Ollama locally on Apple M3 for private, zero-cost LLM inference with cited answer generation —running 100% locally with no external APIs or data leaks.
- ▶ **End-to-End Observability:** Integrated Langfuse to trace every user query, monitor retrieval quality, track latency, and systematically identify pipeline failure points.

EDUCATION

B.Tech, Computer Science · Birla Institute of Technology, Mesra CGPA: 8.68 · 2019 – 2023

Coursework: Distributed Systems, Low-Level Design / Design Patterns, High-Level Design

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), TypeScript, C++, Dart, SQL
Backend & Databases	FastAPI, Node.js, Nest.js, Redis, Celery, PostgreSQL, MongoDB, Qdrant, REST, gRPC, Distributed Systems, WebSockets
AI & Observability	LlamaIndex, Ollama, Langfuse, Docling, BGE-Reranker, phi4, Vector Embeddings, Hybrid Search, RAG Pipelines
Frontend / Mobile	React.js, Redux, Flutter, Riverpod, HTML/CSS, Chrome Extension APIs
Infrastructure & Concepts	AWS (EC2, S3, RDS, Lambda, SQS), Docker, CI/CD, OAuth2.0, RBAC, CRDT